Phishing Email Detection & Awareness Report

Prepared By: Nhletelo Shibambu

Task: Cyber Security Task 2 (Future Interns 2026)

Date: 24 April, 2025

1. Executive Summary

This report presents an analysis of a phishing email and explains how such attacks work. Phishing is one of the most common cyber threats, where attackers try to trick users into sharing sensitive information such as passwords or personal details.

A sample email was analyzed to identify warning signs, classify its risk level, and explain how users can avoid similar attacks.

The findings show that the email contains multiple phishing indicators, including a fake link, urgent messaging, and lack of personalization.

This report also provides simple guidelines to help users recognize and prevent phishing attacks.

2. Objective

The purpose of this report is to:

- Identify phishing indicators in emails
- Analyze how phishing attacks work
- Classify the risk level of the email
- Educate users on how to stay safe
- Provide prevention and awareness guidelines

3. Tools Used

The following tools were used during the analysis:

- Email Header Analyzer – to inspect sender details
- Browser Tools – to check links safely
- Public phishing email samples – for analysis
- Documentation tools (Word / PDF) – for reporting

4. Sample Email Analyzed

Subject: Urgent: Your Account Will Be Locked

Email Body:

Dear User,

We noticed suspicious activity on your account.

To avoid account suspension, please verify your details immediately.

Verify Now: http://secure-account-verify[.]com

Failure to verify within 24 hours will result in permanent account lock.

Regards,
Security Team

Email Header Information

From:	Security Team <support@secure-account-verify.com>
To:	shibambunhletelo@gmail.com
Subject:	⚠️ Urgent: Verify Your Account
Date:	Mon, 20 Apr 2026 10:00:00 +0000
Reply-To:	attacker@gmail.com

Email Content (Body)

Dear User,

We noticed suspicious activity on your account.
To avoid account suspension, please verify your details immediately.

👉 Verify Now: http://secure-account-verify[.]com

Failure to verify within 24 hours will result in permanent account lock.

Regards,
Security Team

Note: This email is a simulated phishing example for educational purposes only. Do not click on any links.

Identified Phishing Indicators

- Sender email uses a suspicious domain (secure-account-verify.com)
- Reply-To address does not match the sender domain
- Uses urgency and fear to trick the user
- Link uses a suspicious URL (http://secure-account-verify[.]com)
- Generic greeting ("Dear User") instead of the recipient's name
- Threatening language about account suspension

Risk Classification

PHISHING
Risk Level: HIGH

Reasoning

The email contains multiple phishing indicators including suspicious sender domain, mismatched reply-to address, urgent/threatening language, and a malicious link. These factors indicate a high risk of phishing attempt.

Recommendation

- Do not click on the link
- Do not provide any personal or account information
- Report the email to your IT or Security Team
- Delete the email immediately

5. Phishing Indicators Identified

The following warning signs were found in the email:

5.1. Generic Greeting

The email uses "Dear User" instead of the recipient's real name.
This shows the email was sent to many people and is not personalized.

5.2. Urgent and Fear-Based Language

The message creates panic by saying the account will be locked within 24 hours.
This is a common tactic used to pressure users into acting quickly without thinking.

5.3. Suspicious Link

The link provided is:
http://secure-account-verify[.]com

This is not an official or trusted domain.
Attackers use fake websites that look real to steal user information.

5.4. Lack of Official Information

The email does not include:

- Company name
- Contact details
- Official branding

This is a strong sign of phishing.

6. Email Analysis

The email is designed to trick users into clicking a malicious link.
Once the user clicks the link, they are redirected to a fake website that looks legitimate.

The fake website then asks the user to enter sensitive information such as:

- Username
- Password
- Banking details

This information is then captured by the attacker and used for fraud or identity theft.

7. Risk Classification

Classification: PHISHING

Risk Level: HIGH

Reason:
The email contains multiple phishing indicators, including a fake link, urgency tactics, and lack of verification.

Users who interact with this email are at high risk of losing sensitive information.

8. How the Attack Works

1. The attacker sends a fake email to the user
2. The email creates urgency or fear
3. The user clicks the malicious link
4. The user enters personal information
5. The attacker steals and uses the data

9. Prevention & Awareness Guidelines

9.1. General Safety Tips

- Do not click on suspicious links
- Always check the sender's email address
- Avoid responding to urgent or threatening emails
- Never share passwords or OTPs

9.2. How to Identify Phishing Emails

- Look for spelling and grammar mistakes
- Check if the message is generic
- Verify links before clicking
- Confirm requests directly with the company

9.3. What To Do If You Receive a Phishing Email

- Do not click any links
- Do not download attachments
- Report the email to your IT department
- Delete the email immediately

10. Do’s and Don’ts

DO:

- Verify sender identity
- Check links carefully
- Report suspicious emails

DON’T:

- Share personal information
- Click unknown links
- Trust urgent requests

11. Conclusion

Phishing attacks are designed to trick users rather than hack systems directly.

This report showed that even a simple email can contain multiple warning signs of an attack. By understanding these signs and following basic safety practices, users can avoid becoming victims of phishing.

Awareness and caution are the most effective ways to prevent phishing attacks.

12. Disclaimer

This report is created for educational purposes only. The phishing email analyzed is a sample used to demonstrate common attack techniques.